

SUMMARY

A lattice is a partially ordered set (Poset) in which every pair of elements has a unique Least Upper Bound (LUB or Join) and a unique Greatest Lower Bound (GLB or Meet). This document presents a structured synthesis of bounded, complemented, distributive, and Boolean lattices, along with sublattice criteria and structural tests for distributivity.

Key insights include:

- A subset qualifies as a sublattice only if it preserves original LUB and GLB operations.
- In a distributive lattice, no element can have more than one complement.
- Lattices with four or fewer elements are always distributive.
- Presence of L1* (pentagon) or L2* (diamond) as a sublattice implies non-distributivity.
- Boolean lattices are complemented and distributive, with size equal to 2^n .

I. FUNDAMENTAL DEFINITIONS AND CORE PROPERTIES

A lattice is a Poset $(L, \leq)$ such that for all $a, b \in L$:

- Join $(a \vee b)$ exists uniquely.
- Meet $(a \wedge b)$ exists uniquely.

Without a partial order, the structure cannot be a lattice.

1.1 Core Algebraic Laws (Always Hold in Any Lattice)

Law Type	Property
Idempotent	$a \vee a = a ; a \wedge a = a$
Commutative	$a \vee b = b \vee a ; a \wedge b = b \wedge a$
Associative	$(a \vee b) \vee c = a \vee (b \vee c)$
Absorption	$a \vee (a \wedge b) = a ; a \wedge (a \vee b) = a$

Important

Distributive law does not hold in all lattices.

1.2 Bounded Lattices

A lattice is bounded if it contains:

Element Type	Description
Minimum (0 or O)	Bottom element
Maximum (1 or I)	Top element

II. COMPLEMENTS IN LATTICES

2.1 Definition of Complement

In a bounded lattice, elements a and b are complements if:

$$a \vee b = 1$$

$$a \wedge b = 0$$

2.2 Rules Governing Complements

Rule Type	Statement
Bound Complement Rule	$0' = 1$ and $1' = 0$
Existence	An element may have 0, 1, or multiple complements
Path Heuristic (Hasse)	Elements on same chain cannot be complements (except 0 and 1)
Totally Ordered Set	Only 0 and 1 can have complements

2.3 Complemented Lattice

A lattice is complemented if:

Every element has at least one complement.

III. SUBLATTICES

A subset M of lattice L is not automatically a sublattice.

3.1 Criteria for Sublattice

Condition	Requirement
Non-Empty	$M \neq \emptyset$
Lattice Structure	M forms a lattice under same relation

Operation Preservation	Join and meet in M equal those in L
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If join or meet differs from original lattice, M is a lattice but not a sublattice.

IV. DISTRIBUTIVE LATTICES

A lattice is distributive if for all a, b, c:

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

4.1 Distributivity vs Complements

Condition	Distributive Status
Any element has more than one complement	Never distributive
Every element has exactly one complement	Definitely distributive
Each element has at most one complement	May or may not be distributive

4.2 Structural Identification Rules

Rule Type	Conclusion
≤ 4 elements	Always distributive
Totally Ordered Set	Always distributive
Contains L1* (Pentagon)	Not distributive
Contains L2* (Diamond)	Not distributive

Important

The structure must be a sublattice isomorphic to L1* or L2*.

A mere visual resemblance is insufficient.

V. BOOLEAN LATTICES

A Boolean lattice is the most structured lattice form.

5.1 Identification Criteria

A lattice is Boolean if and only if:

1. It is distributive.
2. It is complemented.
3. Every element has exactly one complement.

5.2 Structural Properties

Property Type	Formula / Condition
Total Elements	2^n
Hasse Edges	$n \times 2^{n-1}$

5.3 Edge Formula Examples

Number of Vertices	Edge Calculation	Total Edges
2 (2^1)	1×2^0	1
4 (2^2)	2×2^1	4
8 (2^3)	3×2^2	12